



Potent phytoestrogens from *Pyrenacantha volubilis* targeting ER α : a promising natural alternative for breast cancer therapy

Maya Ammathil Manoharan^{1,2} · Senthilkumar Umapathy¹ · Beutline Malgija²

Received: 20 January 2025 / Accepted: 9 April 2025 / Published online: 24 May 2025
© King Abdulaziz City for Science and Technology 2025

Abstract

Selective estrogen receptor modulators (SERMs) play a vital role in ER + ve breast cancer therapy, but existing synthetic drugs have limitations. This study explores *Pyrenacantha volubilis* as a potential natural source for SERM using molecular docking, ADMET profiling, molecular dynamics (MD) simulations, and MM-GBSA analysis, using the Schrödinger Suite. Nine metabolites reported from the species were subjected to analysis, and all except one exhibited superior activity compared to the control. Among these, pumiloside (− 13.06 kcal/mol), strictosidine (− 11.808 kcal/mol), deoxypumiloside (− 11.686 kcal/mol), and strictosamide (− 11.479 kcal/mol) demonstrated stronger ER α binding affinities than control drugs tamoxifen and raloxifene. The receptor–ligand complexes of pumiloside and strictosidine exhibited reasonable interactions, with strictosidine showing the highest glide energy (− 52.418 kcal/mol). MD simulations and MM-GBSA analysis further confirmed the stability of these complexes under physiological conditions, with both compounds showing superior free binding energy compared to tamoxifen. Deoxypumiloside also emerged as a promising candidate with no Lipinski rule of five violations and high oral absorption, but required solubility enhancements. Additionally, 10-hydroxy camptothecin, protonated strictosidine, camptothecin, 9-methoxy camptothecin, 20-deoxy camptothecin, and 20-hexanoyl-10-methoxy camptothecin are other metabolites that exhibited good docking scores and favorable drug-like properties. These findings highlight *P. volubilis* as a promising SERM source, with strictosidine and pumiloside as lead candidates due to their superior receptor interactions, stability, and energetics. Further in vitro and in vivo studies are required to validate their therapeutic potential in breast cancer treatment.

Keywords Estrogen receptor alpha · Breast cancer · *Pyrenacantha volubilis* Wight. · Molecular docking · Molecular dynamics simulation · Network analysis

Introduction

Breast cancer is the leading cause of cancer-related deaths in women worldwide, with 70% of cases driven by estrogen receptor-positive (ER +) signaling pathways. The primary

treatment for ER + breast cancer involves selective estrogen receptor modulators (SERMs) such as tamoxifen, which are effective in reducing tumor growth. However, these agents often exhibit estrogenic effects in non-target tissues, leading to adverse side effects that limit their clinical use. Consequently, attention has shifted toward natural compounds such as genistein, resveratrol, and epigallocatechin-3-gallate (EGCG), which have shown promise in modulating estrogen receptor alpha (ER α) activity. These compounds may provide therapeutic benefits with fewer off-target effects, making them promising candidates for further exploration as adjuncts or alternatives to conventional SERM therapies (Ahmed et al. 2023).

Estrogen receptors, classified as type I nuclear hormone receptors, are integral to the steroid–thyroid–retinoid receptor superfamily and regulate gene expression in response to estrogen (McInerney et al. 1996; Weatherman et al. 1999).

✉ Beutline Malgija
beutline.bioinfo@gmail.com

Maya Ammathil Manoharan
mayaammathilabc12@gmail.com

Senthilkumar Umapathy
sensonsam85@gmail.com; senthilumapathy@mcc.edu.in

¹ DNA Barcoding Laboratory, Department of Botany, Madras Christian College, Chennai 600059, India

² Center for Computational Informatics Research and Innovation, MCC-MRF Innovation Park, Madras Christian College, Chennai 600059, India

ER proteins primarily function as ligand-activated transcriptional regulators, mediating hormonal responses and exist in two primary isoforms, estrogen receptor alpha (ER α) and beta (ER β). ER α , the first discovered steroid receptor (Jensen and Jacobson 1960), is predominantly expressed in breast cancer cells. Both isoforms serve as transcriptional regulators and play key roles in activating multiple kinase signaling pathways. Despite their structural similarity, ER α and ER β often exert opposing physiological effects, adding complexity to estrogen signaling. The ESR1 gene encodes ER α , while ESR2 gene encodes ER β . ER α is a 66 kDa protein consists of 590 amino acids and is organized into six functional domains (A–F). Among these, the A/B domain, also known as the N-terminal domain, plays a crucial role in receptor activation. ER α is a key target in breast cancer therapy due to its high prevalence in mammary tissue, where it is more abundant than ER β . Its activation can occur through ligand binding or phosphorylation, with the latter enabling receptor activation independent of ligand presence. This dual activation mechanism adds complexity to estrogen signaling and underscores ER α 's significance in breast cancer progression and treatment strategies (Anbalagan and Rowan 2015; Ariazi et al. 2006; Muccee et al. 2024) (Fig. 1).

Breast cancer encompasses a range of subtypes, each with distinct molecular characteristics that influence their behavior and response to treatment (Szklarczyk et al. 2019). A key feature of many breast cancer subtypes is the expression of estrogen receptor alpha (ER α), which, when bound by estrogen, triggers a cascade of signaling events that promote tumor growth. ER-positive (ER +) breast cancers are particularly responsive to hormonal therapies that disrupt estrogen signaling, underscoring the critical role of ER α in tumor progression. As a result, targeting ER α remains a central strategy in the management of breast cancer, highlighting the importance of understanding its activation mechanisms for improving therapeutic outcomes (Tecalco-Cruz et al. 2019). Estrogen receptor signaling pathways play a crucial role in regulating key cellular processes such as cell proliferation, cell cycle progression, and apoptosis. As a result, downregulating or inhibiting estrogen receptor activity has become a fundamental therapeutic strategy for treating ER-positive breast cancers (Alzahrani et al. 2024).

Estrogens, the 17 β -estradiol (E2), regulate numerous cellular processes through both genomic and non-genomic mechanisms (Cersosimo and Benarroch 2015; Saczko et al. 2017). This exerts its genomic effects by binding to estrogen receptors, enabling them to function as transcription factors. This binding promotes the transcription of genes involved in cell proliferation, inflammatory responses, and developmental processes. Conversely, estrogen receptors (ERs) also initiate non-genomic mechanisms, by activating pathways such as MAPK, PI3 K, tyrosine kinase cascades, and G-protein-coupled receptors. These non-genomic mechanisms also play a role in cellular responses, independent of direct gene transcription, contributing to diverse physiological outcomes (Saczko et al. 2017; Weatherman et al. 1999). Upon estrogen binding or phosphorylation by cellular kinases, ER α dissociates from heat shock proteins (HSPs) with which it is normally associated. Estrogen receptors can also be activated by other growth factors. When activated, ER α translocates to the nucleus, where it either directly binds to estrogen response elements (EREs) on DNA or associates with transcription factors such as AP1 and SP1 to bind to serum response elements (SREs) on DNA. Additionally, activated ER α interacts with kinases such as PI3 K and Src, subsequently activating signaling pathways like MAPK and AKT, which modulate transcription factors that regulate gene expression (Fig. 2) (Arnal et al. 2017; Girgert et al. 2019). In addition to these mechanisms, ER α are also activated in a ligand-independent manner via phosphorylation, often induced by growth factors such as EGF, which stimulates the MAPK pathway. Many studies have shown that cytokines and other tumor cell-derived growth factors play a significant role in ER-positive breast cancer, where ER activation leads to uncontrolled cell proliferation. Blocking ER receptors with metabolites that target these pathways can have an antagonistic effect, potentially offering therapeutic benefits for ER-positive breast cancer (Roy and Moses 2012). Not only that, breast cancers are also caused by abnormal cell proliferation linked to dysregulated estrogen receptor activity (Barone et al. 2010).

Tamoxifen is a well-known SERM widely used in the treatment of ER-positive breast cancer (Lumachi et al. 2013). Raloxifene, another SERM, shares properties with

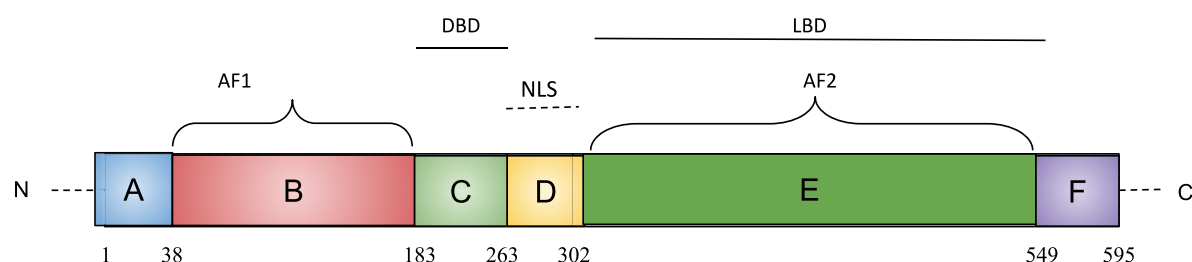


Fig. 1 Structural representation of the ESR1 gene encoding the estrogen receptor alpha (ER α)

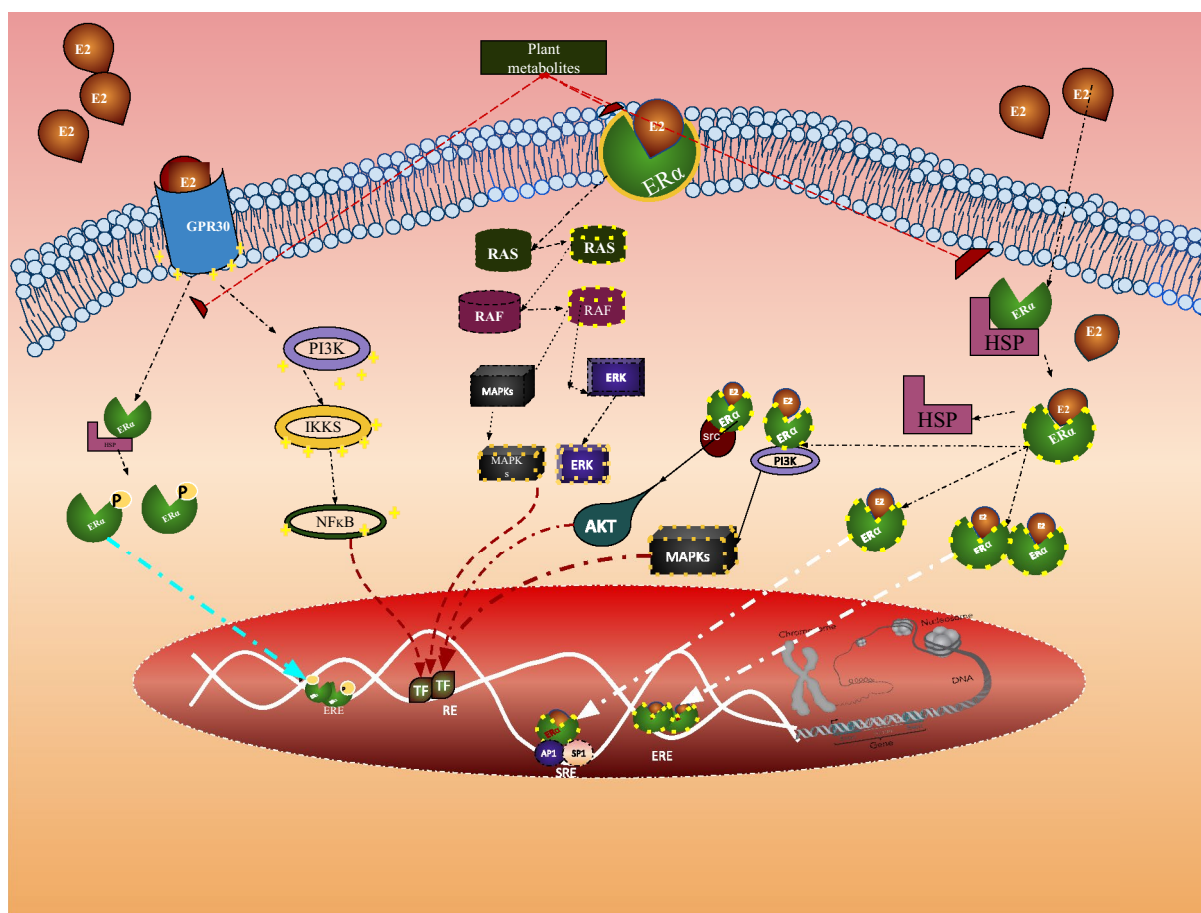


Fig. 2 Estrogen signaling pathway depicting key regulatory elements, including the estrogen response element (ERE), steroid response element (SRE), and response element (RE), which mediate gene transcription upon ligand binding

tamoxifen, with both displaying tissue-specific estrogenic and antiestrogenic effects. In breast cancer cells, tamoxifen functions as an antiestrogen, making it an effective therapeutic option for ER-positive breast cancer (Fisher et al. 1996, 1998). Similarly, raloxifene also acts as an estrogen receptor antagonist and has shown efficacy in breast cancer treatment (Ettinger et al. 1999). While several such synthetic drugs are available for breast cancer treatment, recurrence remains a significant challenge. These treatments, although effective in the early stages, are often expensive. In this context, natural compounds, especially those derived from plants, offer a promising alternative. They not only have the potential to reduce side effects, but also to lower treatment costs. Recent studies have increasingly used molecular docking simulations to investigate the biological properties of plant metabolites, highlighting their potential in drug discovery (Valera et al. 2018).

Several studies have highlighted the anticancer potential of various plant extracts, positioning them as promising candidates for further research. *Pyrenacantha volubilis* is one such plant, demonstrating significant anticancer properties.

Phytochemical studies on *P. volubilis* have reported the presence of metabolites involved in camptothecin (CPT) synthesis, with CPT concentrations being particularly high in the plant's cotyledons. This plant serves as a promising source of bioactive metabolites with potential anticancer properties. While its extracts have shown activity against breast cancer cell lines like MCF-7, the specific metabolites responsible for this effect remain unclear (Suma et al. 2014). Given the critical role of ERα in breast cancer progression, inhibiting this receptor is a key therapeutic strategy. Since *P. volubilis* contains metabolites involved in the post-strictosidine pathway of camptothecin biosynthesis, investigating their potential as a source of SERMs has become essential. To assess their efficacy, molecular docking, ADMET profiling, MD simulations, and MM-GBSA analysis were performed using the Schrödinger Suite, with tamoxifen and raloxifene simultaneously processed as controls. These controls were selected due to their established role in breast cancer therapy, enabling a direct comparison of the binding affinities, pharmacokinetics, and safety profiles of the natural metabolites with clinically approved drugs. The aim of this study is

to identify the most effective metabolite of *P. volubilis* as a natural SERM, offering a potentially safer and more effective alternative for ER α -targeted breast cancer treatment.

Materials and methods

Protein preparation for molecular docking

ER α was investigated as a key receptor in breast cancer due to its role in activating major signaling pathways that drive breast cancer progression. To validate its significance as a therapeutic target, a protein–protein interaction analysis was conducted using the STRING database (<https://string-db.org>) to identify the interacting proteins and their roles in breast cancer-associated pathways [53]. Key regulators of the ER α signaling pathway, including RAS, RAF1, MAPKs, PI3 K, GPR30, HSP90, IKK, NF- κ B, AKT, and SRC, were also considered in the analysis (Clusan et al. 2021; Dimauro et al. 2021). The STRING platform enabled the identification of biological processes and pathways associated with these proteins. STRING's visualization tools were used to construct interaction networks, illustrating key proteins involved in breast cancer signaling mechanisms. The protein interaction network was generated using the K-means legacy clustering algorithm with a confidence threshold of 0.900 (MacQueen 1967). STRING'S integrated experimental data, curated databases, and computational predictions, enhancing the understanding of ER α signaling and reinforcing its potential as an optimal therapeutic target for endocrine therapy in breast cancer (Bader and Hogue 2002).

The X-ray crystallographic structure of ER α (PDB ID: 2IOG), containing a ligand-bound complex within the ligand-binding domain, was retrieved from the RCSB Protein Data Bank (PDB). The associated ligand, identified as IOG, consists of 600 atoms. The retrieved protein structure had a resolution of 1.60 Å, with R-values of 0.328 (R-free) and 0.212. (observed). It comprised a single A chain with 246 residues, with no non-standard residues present. This co-crystallized ligand was selected for grid due to its binding properties and inhibitory activity to identify metabolites capable of modulating ER α signaling. The co-crystallized ligand belonged to the indole class, and exhibited antagonistic activity against estradiol in both uterine and MCF-7 breast cancer cells, supporting the selection of the 2IOG protein structure as a suitable model for evaluating potential ER α inhibitors. This ensured the reliability of the study for evaluating the inhibitory potential of the considered ligands (Dykstra et al. 2007). Structural validation of protein was performed to identify and rectify potential errors. The target receptor was prepared for molecular docking using the Protein Preparation Workflow in Schrödinger software to optimize the structure for docking analysis. The protein was

optimized by adding missing atoms, removing water molecules, and refining the structure (Cao et al. 2013; Valera et al. 2018).

Ligand preparation

Nine metabolites from *P. volubilis*, a plant recognized for its potential as an alternative source of anticancer compounds, including camptothecin, were selected for analysis (Suma et al. 2014). Tamoxifen and raloxifene, two well-established selective estrogen receptor modulators (SERMs) used in breast cancer treatment, were included as control ligands (Nagandla and Thomas 2024; Xu et al. 2024). The metabolites selected for the present study include camptothecin (CPT), 10-hydroxycamptothecin, 9-methoxycamptothecin, 20-hydroxycamptothecin, 20-hexanoyl-10-methoxycamptothecin, pumiloside, deoxypumiloside, strictosidine, and strictosamide (Fig. S1). The three-dimensional (3D) structures of all the above-mentioned metabolites were retrieved from PubChem.

Ligand preparation was performed using the LigPrep module in Schrödinger, generating multiple conformers for each ligand to ensure optimal structural representation for docking. Ionization states were calculated at a physiological pH of 7.2 ± 0.2 to account for biologically relevant protonation states. The ligands were energy minimized using the OPLS3 force field, with all other parameters set to their default values. Both protonated and deprotonated forms of the ligands were generated and included in the docking analysis. Nine *P. volubilis*-derived metabolites conformers and two control drugs, tamoxifen (Karunarathna et al. 2024) and raloxifene (Lee et al. 2024), were prepared as ligands. Following LigPrep processing, a total of 16 ligands were considered: 9 plant-derived metabolites, their protonated forms, and 2 control ligands (Cummings et al. 1999; Jordan 2003; Yao et al. 2013).

Ligand docking

The receptor grid was generated by centering on the ligand-binding site of the prepared protein. Flexible ligand docking was employed using the Ligand Docking module in Schrödinger software (Schrödinger et al. 2024; Ahmad et al. 2024). This approach enables ligand conformational flexibility while preserving the protein structure in a rigid state. The extra precision (XP) docking protocol was employed for enhanced docking precision. The docked complexes were examined through 2D visualization to assess ligand–protein interactions. The results were then compared to the control SERMs, tamoxifen and raloxifene, for assessing each ligand's potential to antagonize estradiol activity. The metabolites considered for the analysis were those that participated in camptothecin biosynthesis, specifically the

substrates in the post-strictosidinic acid pathway. Previous studies on *P. volubilis* extracts have identified it as a rich source of natural bioactive compounds with potential therapeutic efficacy against breast cancer. Therefore, it may serve as a significant natural source of compounds exhibiting selective estrogen receptor modulator (SERM) activity against ER α (Suma et al. 2014).

ADMET analysis

The pharmacokinetic properties and drug-likeness assessments of the ligands were analyzed using QikProp v7.2. This predicts the ADMET (absorption, distribution, metabolism, excretion, and toxicity) properties of the selected ligands based on their three-dimensional (3D) structures (Daoud et al. 2021; Adelusi et al. 2022; Pantaleao et al. 2022). Key physicochemical and pharmacokinetic descriptors relevant to drug development were considered in the analysis. These included molecular weight (130–725 Da), dipole moment (1.0–12.5 D), solvent-accessible surface area (300–1000 Å²), hydrophobic SASA (0–750 Å²), vdW PSA (7.0/200.0 Å²), molecular volume (500–2000 Å³), globularity (0.75–0.95), hydrogen bond donors (0–6), acceptors (2–20), and rotatable bonds (0–15), all of which influence absorption, distribution, and permeability. Pharmacokinetic parameters such as aqueous solubility (QPlogS: – 6.5 to 0.5), blood–brain barrier permeability (QPlogBB: – 3.0 to 1.2), Caco-2 and MDCK permeability (low permeability threshold: < 25 nm/s), skin permeability (QPlogKp), plasma protein binding (QPlogKhsa: – 1.5 to 1.5), and CNS activity (– 2 to 2) were assessed. Human oral absorption was predicted, with values below 25% indicating low gastrointestinal uptake. Drug-likeness was evaluated using Lipinski's rule of five (≤ 4 violations) (Lipinski 2004) and Jorgensen's rule of three (≤ 3 violations), alongside a bioavailability score. Potential toxicity risks, including HERG K⁺ channel inhibition (log IC₅₀ < – 5 indicating cardiotoxicity), oral LD₅₀ (rat), AMES toxicity, hepatotoxicity, and skin sensitization, were analyzed. An overall ADME-compliance score, represented by the #stars parameter, was used to assess pharmacokinetic suitability. This metric indicates how many property descriptors fall beyond the optimal range observed in 95% of known drugs, providing insight into the drug-like potential of the studied compounds (Bake et al. 2025; Feinberg et al. 2020).

Molecular dynamic simulations and MM–GBSA analysis

The protein–ligand complex of ER α and the selected ligands were screened based on least docking score, glide energy, and interactions. To validate their stability in the physiological conditions, these complexes were further evaluated through MD simulations using Schrodinger's Desmond.

MD simulation was done in a system with the TIP3P water model and was neutralized with counter ions in the System Builder panel. The solvent model was minimized, ensuring randomized solvent molecule distribution. Simulations were executed under the NPT ensemble at 330 K and 1.01325 bar for 100 ns. Structural stability and interactions were assessed using key metrics, including root mean square deviation (RMSD), root mean square fluctuation (RMSF), hydrogen bonding, and other intermolecular interactions (Schrodinger et al. 2024).

MD simulations were performed on all metabolites that exhibited least docking scores. Among them, the most stable metabolite under physiological conditions was identified based on root mean square deviation (RMSD), root mean square fluctuation (RMSF), and interaction frequency analysis. Subsequently, molecular mechanics–generalized born surface area (MM–GBSA) calculations were conducted using the PRIME module in the Schrödinger Suite. This analysis was performed on the top-performing metabolites from the MD simulations. Additionally, tamoxifen, a well-established selective estrogen receptor modulator (SERM), was included as a reference standard for comparative evaluation. The MM–GBSA analysis was carried out using the optimized potentials for liquid simulations 3 (OPLS3) force field and the variable-dielectric surface generalized born (VSGB) solvation model to determine the binding free energy and stability of the receptor–ligand complexes.

$$\Delta G_{\text{bind}}(\text{bind free energy}) = E_{\text{complex}} - (E_{\text{protein}} + E_{\text{ligand}})$$

In MM–GBSA analysis, the key binding interactions including hydrogen bonding, van der Waals forces, electrostatic interactions, and lipophilic contacts, were analyzed, providing a detailed assessment of protein–ligand interaction energies. Additionally, Glide Emodel incorporated GlideScore, ligand strain energy, and other parameters to evaluate the overall stability and quality of the docking poses. Lower the ΔG value, the stronger is the binding affinity between the ligand and receptor (Bake et al. 2025; Schrodinger et al. 2024). Overall, this study investigates the potential of *P. volubilis* as a natural source of bioactive metabolites exhibiting selective estrogen receptor modulator (SERM) activity, aimed at supporting endocrine therapy for ER α -positive breast cancer.

Results

Protein–protein interaction analysis using the STRING database identified 133 proteins involved in breast cancer-related signaling pathways, with 66 specifically linked to estrogen signaling (Fig. S2). ESR1 (ER α) exhibited the highest interaction network density, underscoring its central role in breast

cancer progression. Therefore, ESR1 (ER α) was selected as the target receptor for in silico analysis to assess its therapeutic potential in endocrine therapy. The ER α structure (PDB ID: 2IOG) was chosen for its high-resolution X-ray crystallographic structure and the presence of a co-crystallized ligand with strong affinity, selectivity, and inhibitory activity against estrogen receptor (ER) signaling (Dykstra et al. 2007). The binding affinities of these ligand–receptor interactions were assessed after docking based on docking scores, glide energy, and other relevant interactions (Table 1).

Docking analysis revealed that 14 ligands (conformers of 9 metabolites and 2 controls) exhibited significant binding affinity, as indicated by their negative docking scores. Among them, 2 were control drugs, while the remaining 12 were metabolites. Except for one protonated conformer of 10-hydroxycamptothecin, all other metabolites had docking scores below -7 kcal/mol (Graph 1). Pumiloside demonstrated the highest binding affinity (-13.06 kcal/mol), followed by strictosidine (-11.81 kcal/mol). The control drugs, raloxifene and tamoxifen, recorded docking scores of -8.856 kcal/mol and -9.417 kcal/mol, respectively. Several metabolites and their conformers outperformed the control drugs, with all scoring lower than -9.533 kcal/mol, suggesting strong antagonistic activity against estradiol at ER. Notably, 6 of the 12 metabolites achieved docking scores lower than CPT, a well-established anticancer metabolite. Furthermore, 10 of the 12 metabolites outperformed the control drugs in terms of docking scores.

The next level of analysis focused on receptor–ligand interactions, which were visualized through 2D interactions. Each docked complex was analyzed to examine the interaction between the metabolites and ER α . The metabolites engaged in different types of non-covalent interactions, including hydrogen bonding, π – π stacking, and aromatic interactions. Hydrogen bonds play a critical role in stabilizing ligand–protein complexes by enhancing binding strength, while π – π interactions facilitate interactions between electron-rich aromatic systems. Aromatic interactions occur between aromatic rings, contributing to ligand stability. Among these, docking scores served as the primary metric for assessing binding affinity. Among the nine metabolites, pumiloside exhibited the lowest negative docking score (-13.06 kcal/mol), where two hydrogen bonds and one aromatic interaction were with the protein. One hydrogen bond was established with lysine at position 531, while the other formed with glycine at position 521. In these interactions, glycine acted as a hydrogen bond acceptor, and lysine served as the donor. The bond lengths ranged from 2.03 to 2.19 Å, with shorter bond lengths generally indicating stronger interactions (Fig. 3). Strictosidine ranked second, with a docking score of approximately -11.808 kcal/mol. Its interaction with the protein involved hydrogen bonding, where lysine at position 531 acted as a hydrogen

bond acceptor with a bond length of 1.89 Å, while aspartic acid (ASP) at position 351 functioned as a donor with a bond length of 1.72 Å. Additionally, an aromatic hydrogen bond was observed with tryptophan at position 383 (Fig. 4). The next metabolite, deoxypumiloside, ranked third with a docking score of -11.689 kcal/mol and interacted with four amino acid residues of the protein. It formed two hydrogen bonds with ASP at position 351, with bond lengths of 1.79 Å and 2.00 Å, respectively. Additionally, it established a hydrogen bond with tyrosine at position 526, with a bond length of 1.70 Å. Finally, the ligand interacted with glutamic acid (GLU) at position 419 and tryptophan (TRP) at position 383 through aromatic hydrogen bonds (Fig. 5).

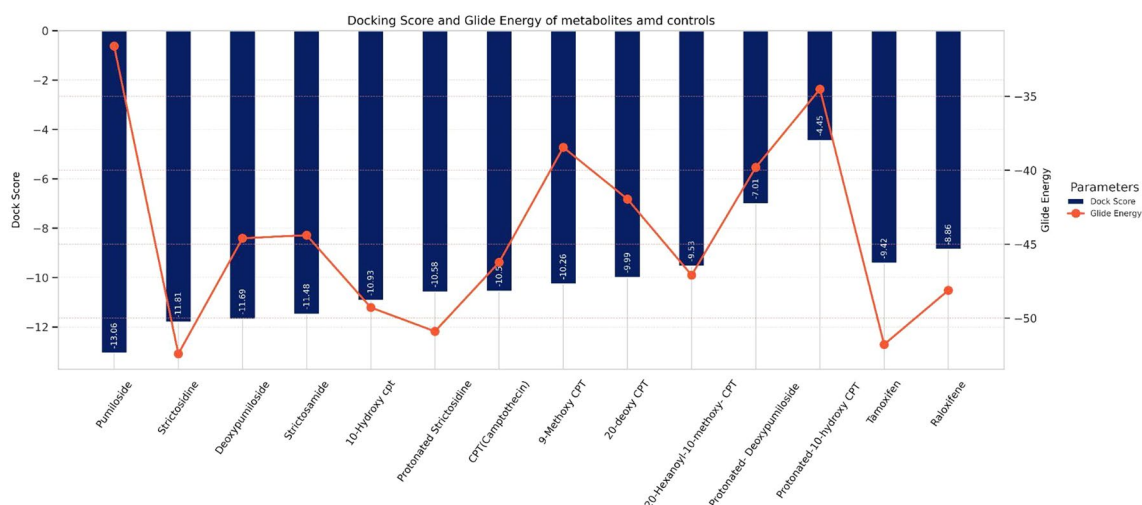
Strictosamide, with a docking score of -11.479 kcal/mol, interacted with three amino acid residues. It formed a hydrogen bond with ASP at position 351, where ASP acted as a hydrogen donor with a bond length of 1.72 Å, and another with tyrosine (TYR) at position 256, which also acted as a hydrogen donor with a bond length of 2.11 Å. Additionally, TRP at position 383 formed an aromatic hydrogen bond with the ligand (Fig. 6).

10-hydroxycamptothecin exhibited a docking score of -10.925 kcal/mol and interacted with two amino acid residues. It formed one hydrogen bond with GLU at position 353, where GLU acted as a hydrogen acceptor, and established two π – π interactions with phenylalanine (PHE) at position 404 (Fig. S3).

The next ranked ligand was a protonated conformer of strictosidine, which displayed a docking score of -10.582 kcal/mol. It interacted with lysine (LYS) at position 531 and aspartic acid (ASP) at position 351 through hydrogen bonds, with both amino acids acting as hydrogen acceptors. An aromatic hydrogen bond was also observed with tryptophan (TRP) at position 383 (Fig. S4). CPT, with a docking score of -10.553 kcal/mol, interacted with phenylalanine (PHE) at position 404 through aromatic interactions, as well as with glutamic acid (GLU) at position 353 and leucine (LEU) at position 387 via aromatic hydrogen bonds (Fig. S5). 9-methoxycamptothecin docked to the protein with a docking score of -10.264 kcal/mol, forming an aromatic hydrogen bond with PHE at position 404 (Fig. S6). The ligand 20-deoxycamptothecin exhibited a docking score of -9.995 kcal/mol, interacting with LEU at position 387 through aromatic interactions and with PHE at position 404 via π – π interactions (Fig. S7). 20-Hexanoyl-10-methoxycamptothecin formed aromatic hydrogen bonds with LEU at position 346 and GLU at position 353. Additionally, it established two aromatic interactions with PHE at position 404 (Fig. S8). The synthetic SERMs, tamoxifen and raloxifene, used as controls, had docking scores of -9.147 kcal/mol and -8.856 kcal/mol, respectively. Tamoxifen formed a π interaction with PHE at position 404 (Fig. S9), while raloxifene established

Table 1 Molecular Docking Profiling and Pharmacokinetic Properties of *Pyrenacantha volubilis* Metabolites and Controls : Key Parameters Including Dock Score, Glide Energy, Hydrogen Bond Interactions, Molecular Weight, QPlogPo/w, Oral Absorption, and Lipinski's Rule of Five.

Ligands	Dock score	Glide score	Glide energy	Molecular Weight (MW)	QPlogP o/w	Hydrogen Bond Donors (HBD)	Hydrogen Bond Acceptors (HBA)	QPPCa co S	QPlog S	Percent Human Oral Absorption	QPlog BB	Lipinski's Rule of Five:	QPlog HERG	QPlog Kp
1 Pumiloside	- 13.06	- 13.06	- 31.604	512.515	- 0.608	5	17.4	25.582	- 3.699	22.669	- 2.866	2	- 6.131	- 4.946
2 Strictosidine	- 11.808	- 13.412	- 52.418	530.574	0.459	6	15.4	15.661	- 3.522	12.14	- 2.382	3	- 7.351	- 5.914
3 Deoxypumiloside	- 11.686	- 11.769	- 44.598	496.516	0.35	4	15.9	85.171	- 3.839	63.544	- 2.247	0	- 6.237	- 3.804
4 Strictosamide	- 11.479	- 11.479	- 44.392	498.532	0.588	5	14.9	72.969	- 4.162	63.735	- 2.308	0	- 6.035	- 4.054
5 10-hydroxy-CPT	- 10.925	- 10.934	- 49.283	364.357	1.179	2	8.5	136.936	- 3.651	72.088	- 1.43	0	- 4.945	- 4.126
6 Protonated Strictosidine	- 10.582	- 10.623	- 50.895	530.574	0.459	6	15.4	15.661	- 3.522	12.14	- 2.382	3	- 7.351	- 5.914
7 CPT(Camptothecin)	- 10.553	- 10.553	- 46.226	348.357	1.742	1	7.75	451.516	- 3.679	84.659	- 0.849	0	- 5.067	- 3.066
8 9-methoxy-CPT	- 10.264	- 10.264	- 38.451	378.384	1.902	1	8.5	443.376	- 3.99	85.453	- 0.977	0	- 5.141	- 3.156
9 20-deoxy-CPT	- 9.995	- 9.995	- 41.951	332.358	2.037	0	7	877.278	- 3.283	91.549	- 0.52	0	- 5.164	- 2.575
10 20-hexanoyl-10-methoxy-CPT	- 9.533	- 9.533	- 47.093	460.529	3.224	0	9.75	689.125	- 4.837	96.621	- 1.191	0	- 5.922	- 2.465
11 Protonated deoxypumiloside	- 7.006	- 8.213	- 39.802	496.516	0.35	4	15.9	85.171	- 3.839	63.544	- 2.247	0	- 6.237	- 3.804
12 Protonated 10-hydroxy-CPT	- 4.451	- 6.936	- 34.511	364.357	1.179	2	8.5	136.936	- 3.651	72.088	- 1.43	0	- 4.945	- 4.126
13 Tamoxifen	- 9.417	- 11.31	- 51.785	371.521	6.381	0	2.75	2133.67	- 5.662	100	0.351	1	- 7.333	- 1.401
14 Raloxifene	- 8.856	- 11.169	- 48.125	473.586	4.792	2	6.25	112.08	- 6.299	91.684	- 1.033	0	- 7.837	- 4.086



Graph 1 Comparison of docking scores and glide energy values of *Pyrenacantha volubilis* metabolites and controls bound to estrogen receptor alpha (ER α), highlighting the binding efficiency and interaction strength

two π - π interactions with the same residue and a hydrogen bond with GLU at position 353 (Fig. S10). In comparison, *P. volubilis* metabolites, namely pumlilide (− 13.06

kcal/mol), strictosidine (− 11.81 kcal/mol), deoxypumlilide (− 11.69 kcal/mol), strictosamide (− 11.47 kcal/mol), camptothecin (− 10.55 kcal/mol), 9-methoxy

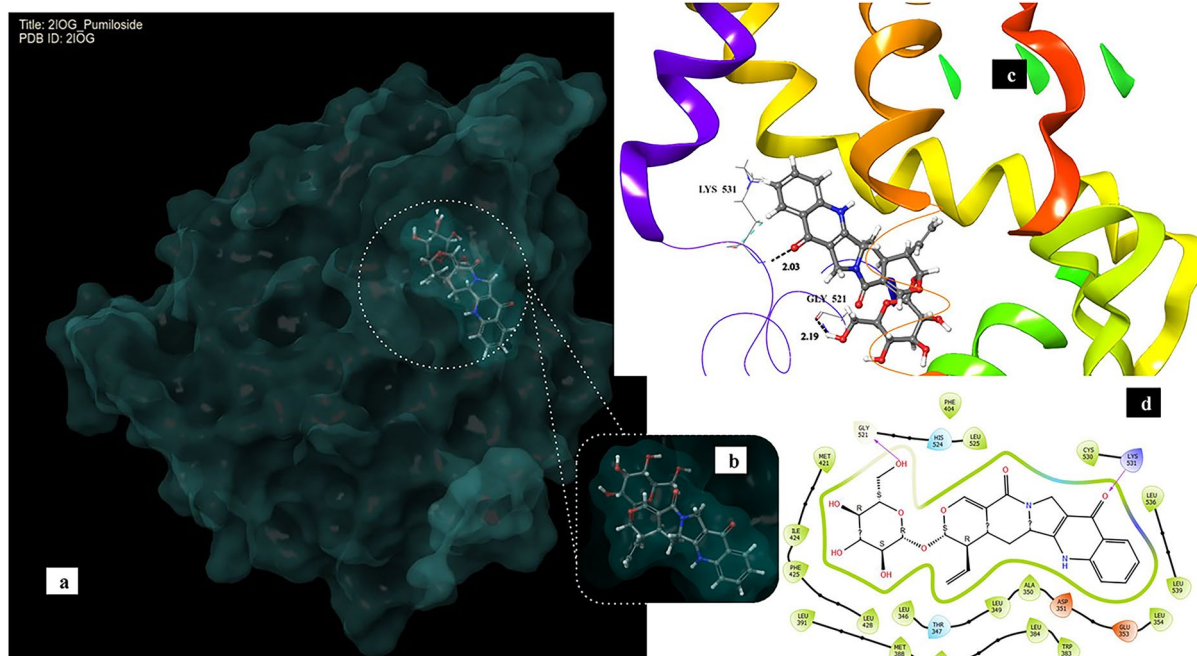


Fig. 3 3D and 2D visualization of pumlilide docked with ER α (PDB ID: 2IOG). **a** 3D surface representation of the ER α -pumlilide complex, illustrating the binding pocket. **b** Chemical structure of pumlilide. **c** 3D visualization of pumlilide interactions with key amino

acid residues of ER α active site. **d** 2D interaction map showing the molecular interactions between pumlilide and ER α residues. Legends: black dashed lines—hydrogen bonds; light blue dashed lines—Van der Waals interactions

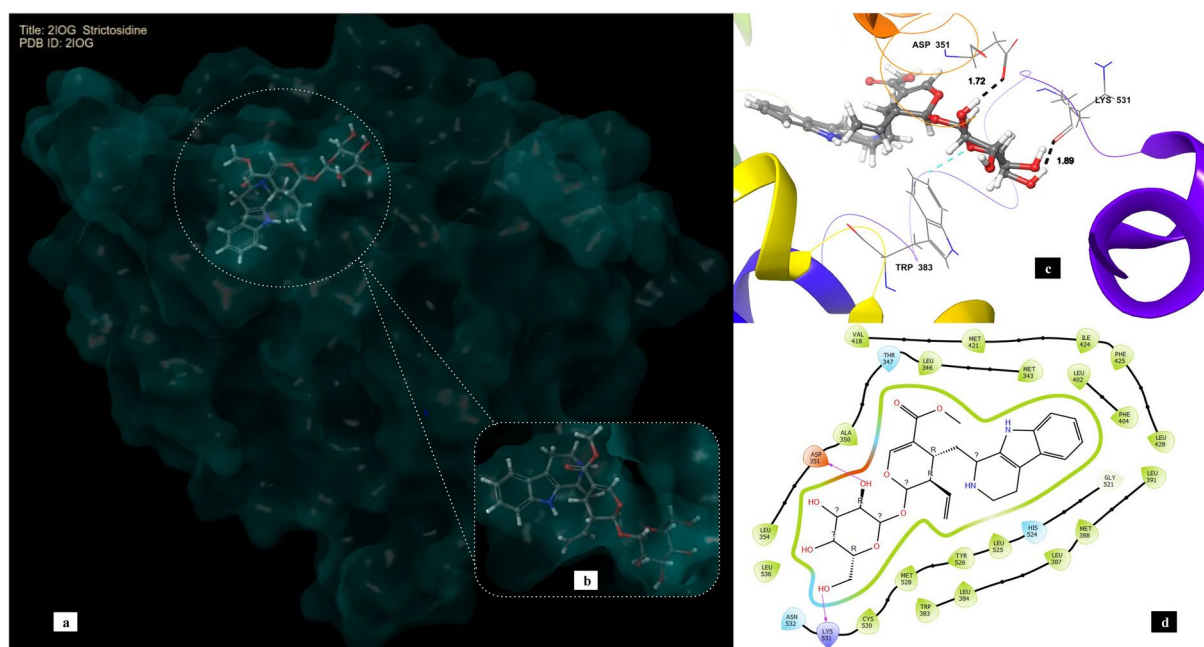


Fig. 4 3D and 2D visualization of strictosidine docked with ER α (PDB ID: 2IOG). **a** 3D surface representation of the ER α -strictosidine complex, illustrating the binding pocket. **b** Chemical structure of strictosidine. **c** 3D visualization of strictosidine interactions with

the key amino acid residues of ER α active site. **d** 2D visualization of strictosidine interactions with the key amino acid residues of ER α active site. Legends: black dashed lines—hydrogen bonds; light blue dashed lines—Van der Waals interactions

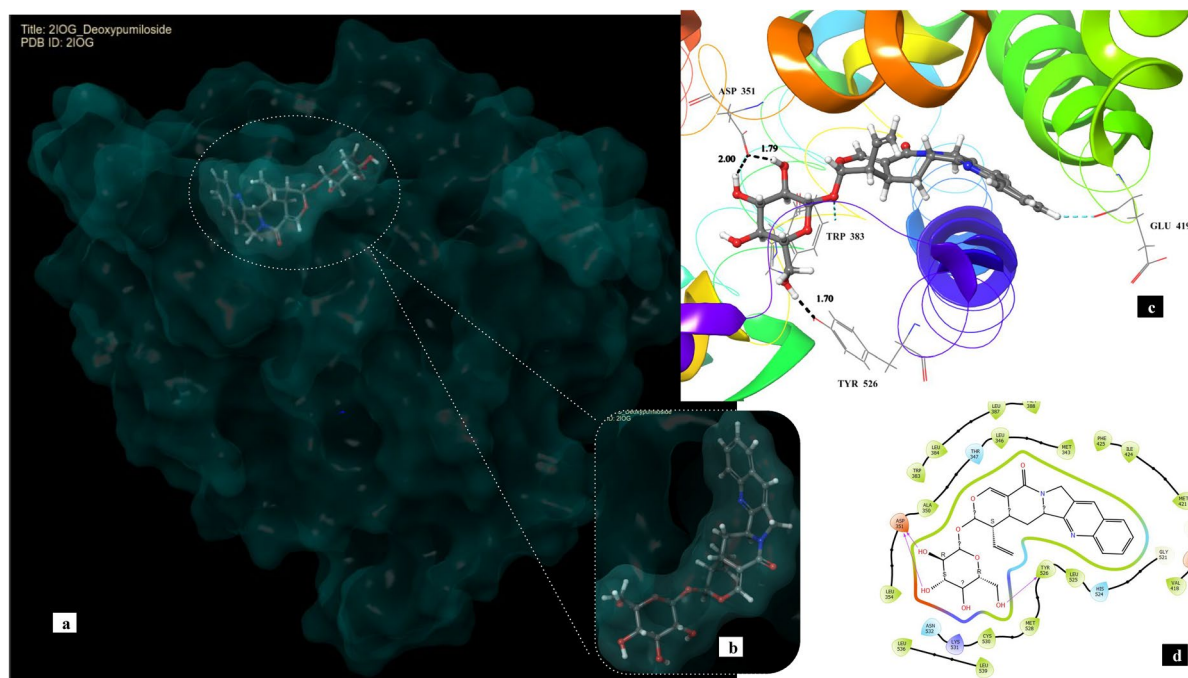


Fig. 5 3D and 2D visualization of deoxypumiloside docked with ER α (PDB ID: 2IOG). **a** 3D surface representation of the ER α -deoxypumiloside complex, illustrating the binding pocket. **b** Chemical structure of deoxypumiloside. **c** 3D visualization of deoxypumiloside's molec-

ular interactions with key amino acid residues of ER α active site. **d** 2D interaction diagram illustrating key interactions with key amino acid residues of ER α active site. Legends: black dashed lines—hydrogen bonds; light blue dashed lines—Van der Waals interactions

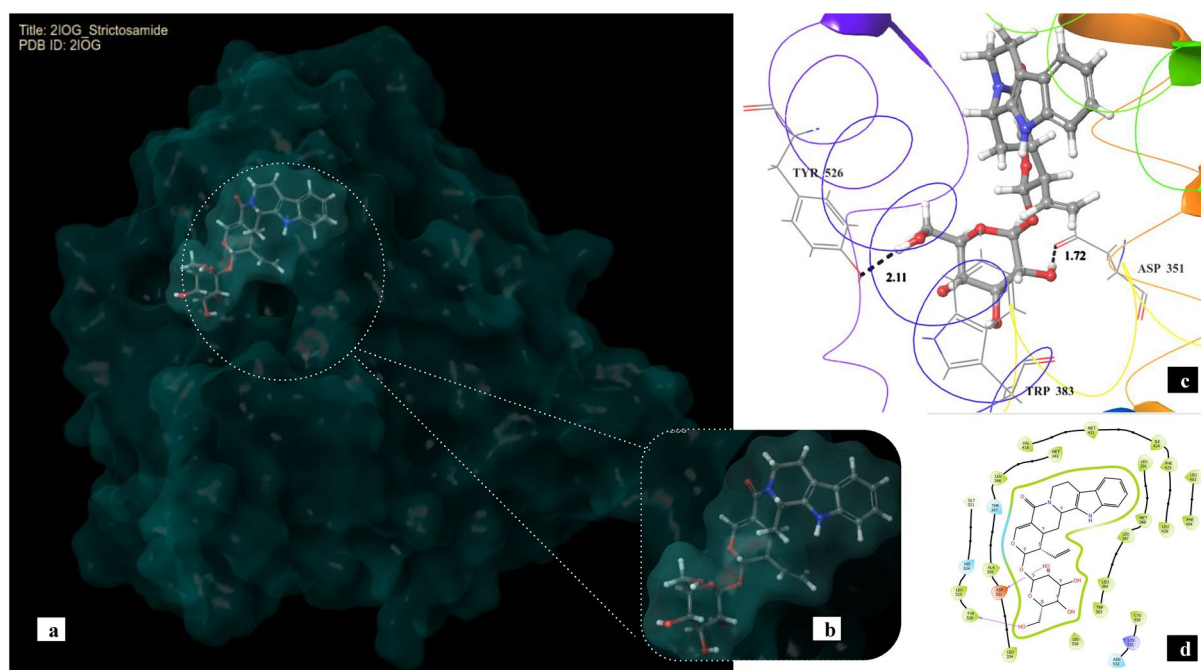


Fig. 6 3D and 2D visualization of strictosamide docked with ER α (PDB ID: 2IOG). **a** 3D representation of the ER α –strictosamide complex, illustrating the binding pocket. **b** Chemical structure of strictosamide. **c** 3D visualization of strictosamide's interactions with key

amino acid residues of ER α active site. **d** 2D interaction map displaying interactions with key amino acid residues of ER α active site. Legends: black dashed lines—hydrogen bonds; light blue dashed lines—Van der Waals interactions

camptothecin (− 10.26 kcal/mol), and 20-deoxy camptothecin (− 9.99 kcal/mol) exhibited stronger ER α interactions, surpassing the control drugs in docking scores, indicating their potential as good ER α modulators. Conversely, protonated deoxypumilosite (− 7.00 kcal/mol) and protonated 10-hydroxy camptothecin (− 4.45 kcal/mol) exhibited weaker binding. protonated deoxypumilosite formed hydrogen bonds with ASP at position 351 and GLU at position 419, while TRP at position 383, GLY at position 420, and GLU at position 419 participated in aromatic hydrogen bonding. Overall, except for one low-affinity metabolite, the remaining ligands demonstrated strong ER α interactions, highlighting their potential as promising SERM candidates for further therapeutic evaluation. Strictosidine exhibited the lowest glide energy (− 52.418 kcal/mol) and formed numerous hydrogen bonds, ionic interactions, and water-mediated bonds with the receptor, surpassing tamoxifen (− 51.785 kcal/mol) and raloxifene (− 48.125 kcal/mol) in both binding strength and interaction count. Based on the docking scores, ten conformers of the metabolites outperformed the control, among which strictosidine (− 13.412 kcal/mol) and pumilosite (− 13.06 kcal/mol) demonstrated superior binding affinities.

ADMET analysis

The docking and ADMET (absorption, distribution, metabolism, excretion, and toxicity) predictions indicated that the selected metabolites showed strong binding affinities to ER α , along with favorable drug-likeness and pharmacokinetic profiles, suggesting their potential therapeutic efficacy in breast cancer treatment (Fig. 7). Evaluation of Lipinski's rule of five for the selected ligands revealed that most of the tested metabolites, including strictosamide, 10-hydroxy camptothecin, protonated strictosidine, 9-methoxy camptothecin, 20-deoxy camptothecin, 20-hexanoyl camptothecin, 20-hexanoyl-10-methoxy camptothecin, and deoxypumilosite, complied with Lipinski's criteria and demonstrated higher docking scores compared to the control drugs, suggesting their potential as promising therapeutic candidates (Fig. 7, Table 1). Strictosidine (three violations) and pumilosite (two violations) showed minor deviations from Lipinski's rules, but remained within the acceptable range. However, these violations could potentially affect membrane permeability and absorption. Since these violations fall within the acceptable limits and all other properties support their therapeutic potential, these metabolites remain strong candidates, albeit with some limitations regarding their oral bioavailability.

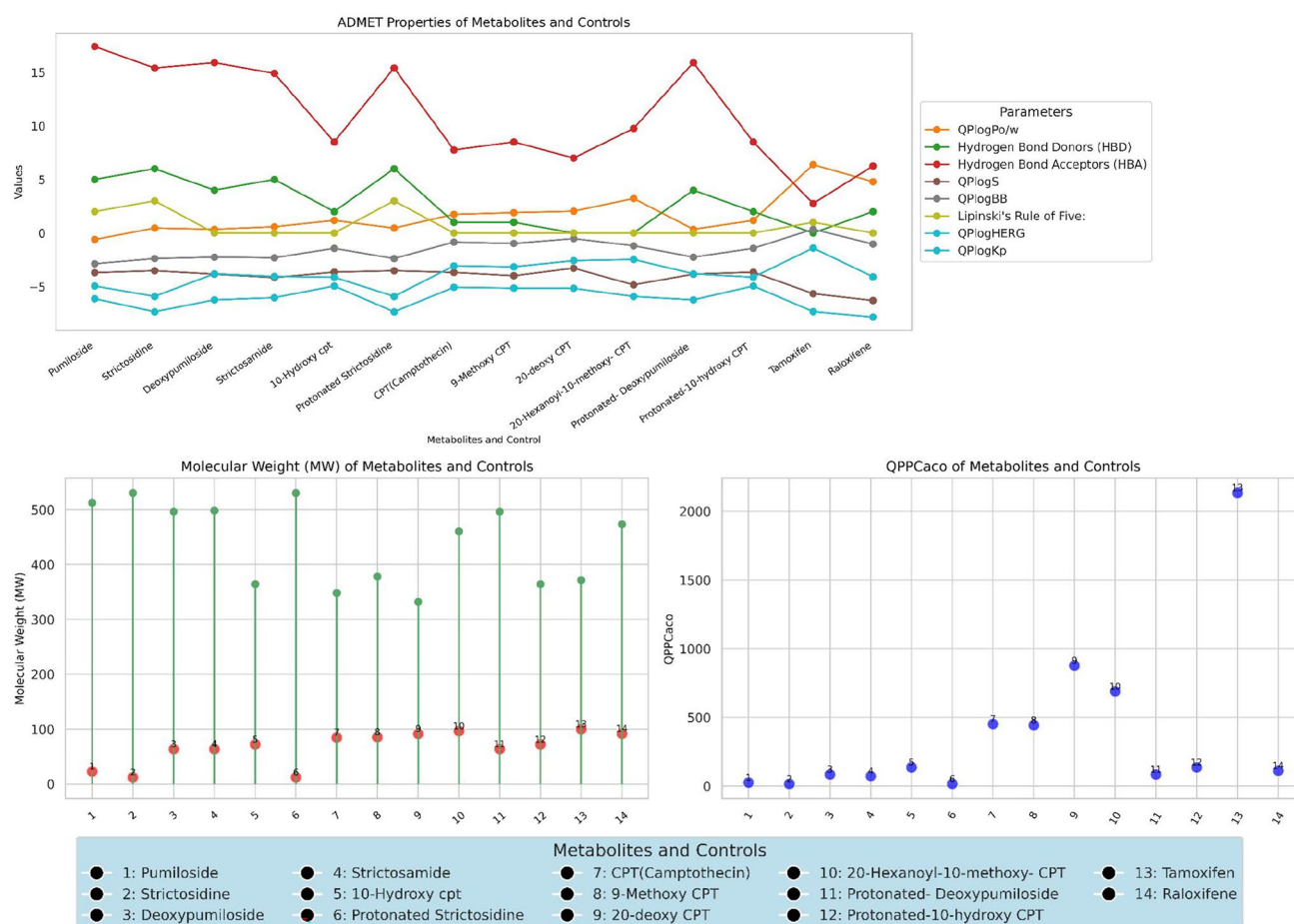


Fig. 7 ADMET profiling of *Pyrenacantha volubilis* metabolites and controls, using key parameters analyzed through the QikProp module in Schrödinger

In contrast, deoxypumiloside fully complied with Lipinski's criteria, exhibiting high oral absorption (63.54%), positioning it as a superior candidate over the control drugs in terms of oral permeability. However, its docking score, glide score, and glide energy were lower than those of strictosidine and pumiloside, suggesting that while deoxypumiloside possesses favorable pharmacokinetic properties, its receptor binding affinity is relatively weaker.

CPT and the remaining metabolites, despite their strong binding scores, exhibited low aqueous solubility, which could impact formulation stability. However, targeted structural modifications may enhance their pharmacokinetic properties, while maintaining receptor binding efficiency. Permeability analysis, assessed using QPPCaco, indicated that tamoxifen (2133.68 nm/s) exhibited high intestinal permeability, contributing to its well-established oral bioavailability. Several *Pyrenacantha*-derived metabolites, including 20-deoxy camptothecin (QPPCaco 877.28) and 9-methoxy camptothecin (QPPCaco 443.38), also demonstrated high

permeability, further supporting their potential for oral drug development.

The blood–brain barrier (BBB) penetration (QLogBB) was highest for tamoxifen (0.351), while most metabolites exhibited moderate-to-low BBB permeability, ranging from -1.0 to -2.8 . Limited BBB permeability is beneficial for SERM drugs as it minimizes the risk of central nervous system (CNS) side effects, thereby improving their overall safety profile (Curtaz et al. 2022).

QLogHERG, a computational descriptor in QikProp (Schrödinger), estimates a compound's likelihood of inhibiting the hERG potassium channel, a key factor associated with QT prolongation and cardiotoxicity (Tie et al. 2009). More negative values indicate higher hERG inhibition, which is undesirable as it increases the risk of cardiotoxicity. Tamoxifen (-7.33) and raloxifene (-7.83) exhibited a high risk of hERG K⁺ channel inhibition, consistent with their known cardiotoxic effects. Similarly, strictosidine (-7.35) displayed significant hERG inhibition, indicating a need for structural refinements to enhance safety. In contrast, most

Pyrenacantha-derived metabolites, including pumiloside (− 6.131) and deoxypumiloside (− 6.237), demonstrated moderate hERG inhibition, suggesting a potentially safer cardiac profile compared to the control drugs. These findings highlight the potential for formulation modifications and metabolic refinements to further reduce cardiac risks, while preserving therapeutic efficacy.

Molecular dynamics simulation analysis

Metabolites with docking scores lower than the control drug were selected for MD simulations. Post-MD simulations, RMSD analysis was conducted from the trajectory over a 100-ns period to evaluate protein–ligand complex stability (Fig. 8). Pumiloside maintained consistent binding throughout, demonstrating high structural stability. Strictosidine also exhibited strong stability, with brief fluctuations of 1–2 Å lasting less than 10 ns. In contrast, strictosamide dissociated from the protein for approximately 80 ns, while 9-methoxycamptothecin displayed deviations of around 3 Å for nearly 50 ns. Other metabolites showed transient deviations, generally within 2 Å for durations under 30 ns, indicating stable binding. Overall, pumiloside and strictosidine exhibited superior stability, reinforcing their potential as SERMs.

Similarly, RMSF analysis provided insights into protein flexibility and identified residue-specific changes upon ligand binding (Fig. 9). The highest fluctuations were observed in the 153–162 residue region of the estrogen receptor (ER) protein, particularly when bound to

strictosamide, 10-hydroxycamptothecin, camptothecin (CPT), and protonated deoxypumiloside. While minor structural variations were detected, RMSF values remained below 4 Å, showing minimal deviation from the unbound protein. A detailed assessment of intermolecular interactions was conducted to evaluate the stability of these metabolites. Key factors influencing stability included hydrogen bonds, ionic bonds, water bridges, and hydrophobic interactions (Fig. S11, S12 and S13). Among the protein–ligand complexes, the ER α –strictosidine complex exhibited all these interactions and contained the highest number of ionic bonds. Pumiloside had the greatest number of hydrogen bond acceptors, while protonated strictosidine displayed the highest number of hydrogen bond donors. Stability analysis ranked strictosidine as the most stable metabolite due to its extensive ionic and non-covalent interactions. The combined findings from RMSF and RMSD analyses further support that pumiloside and strictosidine are the most stable and promising SERMs for ER receptor regulation, consistent with their favorable docking scores and glide energy.

Compared to the control drugs, metabolites strictosidine, pumiloside, deoxypumiloside, and strictosamide demonstrated superior glide scores, highlighting their potential as ER α modulators (Table 1). Additionally, protonated strictosidine, 10-hydroxycamptothecin, and 20-hexanoyl-10-methoxycamptothecin showed competitive glide energy values, further supporting their stability as potential SERMs.

Pumiloside and strictosidine exhibited the most stable interactions with ER α , showing minimal structural fluctuations under physiological conditions. Molecular dynamics (MD) simulations further confirmed that the stability of these metabolites under physiological conditions was

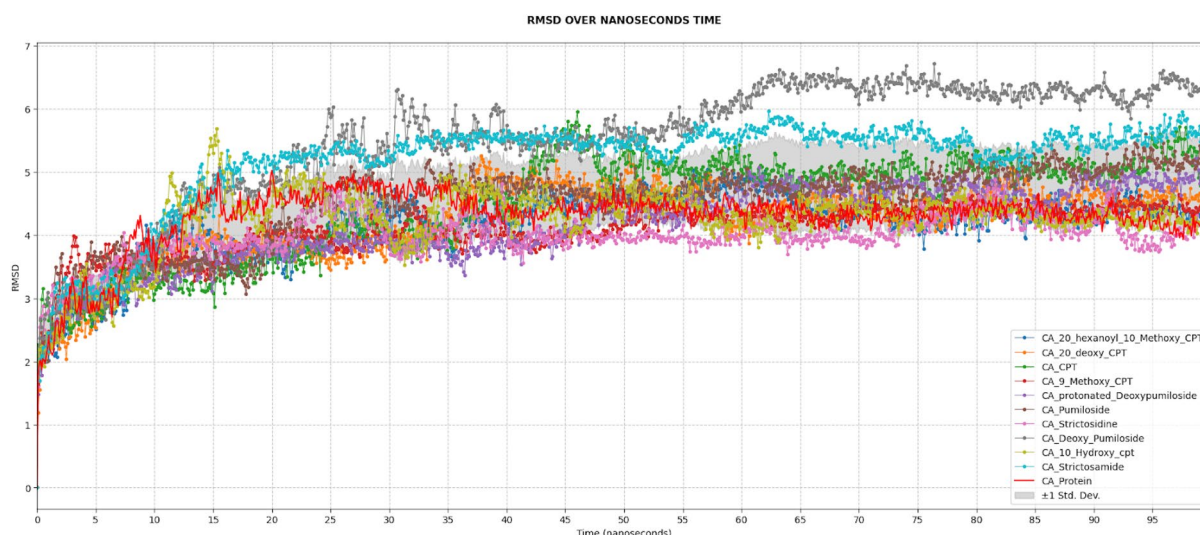


Fig. 8 RMSD (root mean square deviation) analysis over 100 ns molecular dynamics simulations, assessing the structural stability of ER α –ligand complexes of metabolites with high docking scores

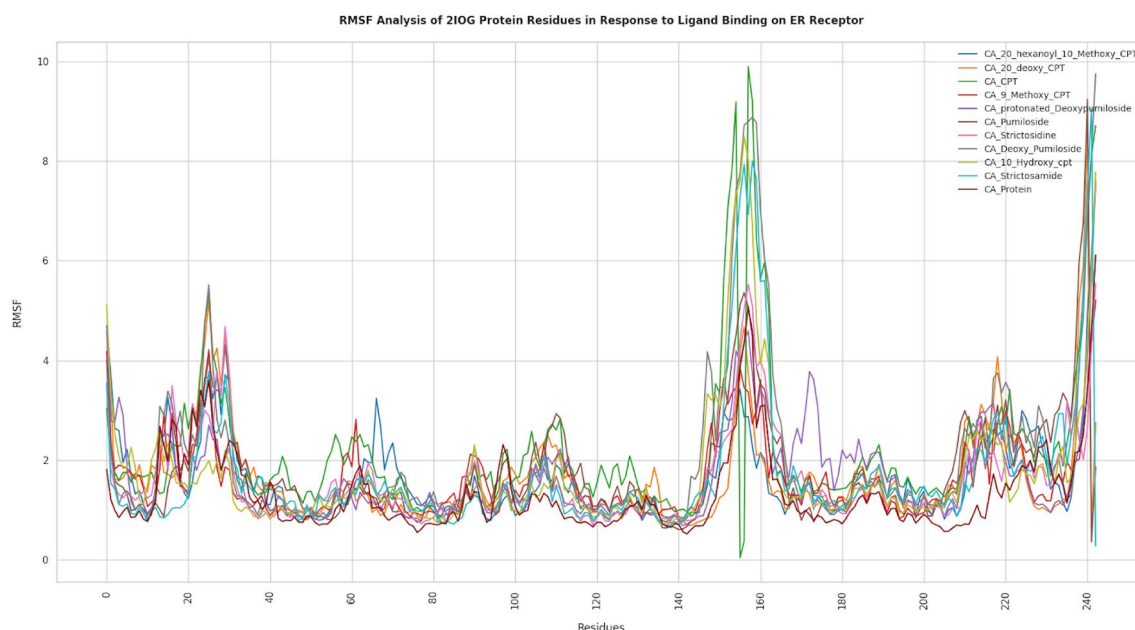


Fig. 9 RMSF (root mean square fluctuation) analysis of the protein upon binding of the selected metabolites from *Pyrenacantha volubilis* over 100 ns simulations, evaluating residue-level flexibility within the binding site

higher than all other tested compounds, suggesting their potential for sustained receptor modulation. In contrast, CPT, despite its established anticancer properties, exhibited lower stability than both pumiloside and strictosidine in the RMSF analysis.

The MM-GBSA (Table 2) analysis revealed that pumiloside exhibits the strongest binding affinity ($\Delta G_{\text{Bind}} = -133.445$ kcal/mol) among the tested compounds, followed by strictosidine ($\Delta G_{\text{Bind}} = -126.539$ kcal/mol) and tamoxifen ($\Delta G_{\text{Bind}} = -113.433$ kcal/mol), which serves as the control. The superior binding affinity of pumiloside is primarily driven by its higher lipophilic contribution ($\Delta G_{\text{Bind lipophilic}} = -82.1804$ kcal/mol), which is greater than that of strictosidine (-70.7083 kcal/mol) and tamoxifen (-70.1565 kcal/mol), indicating stronger hydrophobic interactions within the binding pocket.

Electrostatic interactions ($\Delta G_{\text{Bind Coulomb}}$) also play a crucial role in ligand stability. Strictosidine exhibits the

strongest electrostatic interactions (-20.7116 kcal/mol), followed by pumiloside (-14.8965 kcal/mol), while tamoxifen (-2.07039 kcal/mol) contributes the least. This suggests that strictosidine benefits from stronger electrostatic stabilization, although pumiloside compensates with enhanced lipophilic and van der Waals interactions.

In terms of van der Waals interactions, pumiloside (-64.777 kcal/mol) exhibits the most favorable contribution, surpassing strictosidine (-58.5337 kcal/mol) and tamoxifen (-61.2566 kcal/mol). The solvation energy penalty ($\Delta G_{\text{Bind Solv GB}}$) is lowest for tamoxifen (17.02228 kcal/mol), followed by strictosidine (23.47217 kcal/mol) and pumiloside (24.94763 kcal/mol). Although solvation energy generally opposes binding, its effect is outweighed by the stronger binding interactions observed in pumiloside and strictosidine, compared to tamoxifen.

While MM-GBSA analysis suggests that pumiloside has the most favorable binding profile, MD simulation-based 2D interaction analysis reveals that strictosidine forms more

Table 2 MM-GBSA analysis of Pumiloside, and Strictosidine, plant-derived metabolites, in comparison to Tamoxifen, a synthetic SERM

Receptor ligand complex	ΔG_{Bind} (kcal/mol)	$\Delta G_{\text{Bind Coulomb}}$ (Kcal/mol)	$\Delta G_{\text{Bind Covalent}}$ (Kcal/mol)	$\Delta G_{\text{Bind H Bond}}$ (Kcal/mol)	$\Delta G_{\text{Bind Lipophilic}}$ (Kcal/mol)	$\Delta G_{\text{Bind Solv GB}}$	$\Delta G_{\text{Bind-vdw}}$ (Kcal/mol)
ER α -Tamoxifene	-113.433	-2.07039	3.259295	-0.00013	-70.1565	17.02228	-61.2566
ER α -Strictosidine	-126.539	-20.7116	0.513285	-0.57093	-70.7083	23.47217	-58.5337
ER α -Pumiloside	-133.445	-14.8965	4.265843	-0.14615	-82.1804	24.94763	-64.777

hydrogen bonds with the receptor compared to both pumiloside and tamoxifen, which may contribute to additional binding stability. Furthermore, glide energy analysis indicates that strictosidine exhibits a more favorable binding energy than tamoxifen, reinforcing its potential as an effective SERM (Bake et al. 2025).

The higher docking scores of several metabolites derived from *P. volubilis* suggest a stronger binding affinity to ER α compared to synthetic alternatives, indicating their potential therapeutic activity. Additionally, these metabolites exhibited favorable ADMET properties and structural characteristics that make them promising ER α inhibitors, surpassing the control drugs. While some compounds showed minor Lipinski violations and moderate cardiotoxicity risks, these could be mitigated through strategic structural modifications without affecting receptor binding.

Discussion

Bioactive compounds from natural sources play a crucial role in drug discovery, offering structural complexity, biological compatibility, and high therapeutic potential. Their intricate molecular frameworks enhance target specificity, bioavailability, and safety, often surpassing synthetic alternatives. The evolutionary adaptation of these compounds enables effective interactions with biological systems, making them valuable candidates for novel drug development. Given these advantages, further exploration and optimization of bioactive compounds remain essential for advancing pharmaceutical research (Mishra et al. 2023a, 2023b). Recent advancements in synthetic libraries, screening technologies, metabolic engineering, and synthetic biology have renewed interest in natural metabolites. Enhanced genetic sequencing and pathway engineering now allow for better exploration and modification of these compounds, creating new opportunities for developing safe, effective, and sustainable therapeutics (Gowtham et al. 2024; Li and Vederas 2009). In parallel, *in silico* analyses have become essential tools in drug discovery, offering predictive insights that guide targeted treatment strategies. With continuous advancements in computational methods, these approaches play a growing role in identifying and optimizing therapeutic candidates. This progress underscores the potential of phytochemicals to enhance clinical interventions and contribute to the development of more effective treatments (Bajorath 2002).

Recent studies in cancer research have placed greater emphasis on the development of anti-estrogens aimed at blocking estrogen-driven tumor progression. In this context, natural compounds and their derivatives, which selectively bind to ER α , offer a promising alternative to conventional treatments by potentially overcoming drug resistance and reducing side effects (Sharma et al. 2018).

This study aimed to identify potential lead molecules for breast cancer therapy by inhibiting estrogen receptor signaling. *In silico* docking studies were conducted on nine secondary metabolites from *P. volubilis* to evaluate their binding potential. Metabolites with more negative docking scores indicated stronger binding affinities. Comparable studies on *Artocarpus* species targeting the ER α protein (PDB ID: 2IOG) have demonstrated the therapeutic potential of specific metabolites, such as Artonin E, which exhibited a docking score of -12.72 kcal/mol. These findings further support the potential of plant-derived metabolites in developing novel SERMs for breast cancer therapy (Etti et al. 2016). In the present study, metabolites derived from *P. volubilis* showed even more promising results, with pumiloside achieving the docking score of -13.06 kcal/mol, surpassing tamoxifen the control. Notably, seven of the nine metabolites exhibited superior docking scores compared to the standard drugs tamoxifen and raloxifene, suggesting their potential as SERMs and viable alternatives to tamoxifen in breast cancer treatment. These findings suggest that *P. volubilis* metabolites warrant further investigation as potential therapeutic agents for ER + ve breast cancer.

Existing literature suggests that natural products can modulate ER signaling, disrupt receptor activity, and induce ER stress, ultimately leading to apoptosis. This highlights their potential as therapeutic agents in estrogen-dependent breast cancer treatment (Kim and Kim 2018). Such mechanisms can enhance the effectiveness of breast cancer treatment strategies by leveraging the dual actions of these metabolites. Approximately 70% of estrogen receptor-positive breast cancers initially respond to anti-estrogen therapies, such as SERMs like tamoxifen. However, many patients develop resistance to these therapies over time, reducing their effectiveness in the later stages of treatment. In this context, alternative strategies that involve the inhibition of ER α signaling using plant-derived metabolites offer promising candidates to overcome resistance and improve treatment outcomes (Ali and Coombes 2000). Most of the metabolites from *P. volubilis* targeted ER α and inhibited key cancer-associated pathways, including MAPK, NF- κ B, and AKT. Among these metabolites, pumiloside exhibited the highest docking score, followed by strictosidine, with all eight metabolites showing strong binding potential. Notably, strictosidine stood out for its superior stability, high glide energy than control, suggesting its effectiveness in modulating ER signaling and inhibiting breast cancer cell malignancy. Acting similarly to SERMs, these metabolites demonstrate strong therapeutic potential for estrogen receptor-positive breast cancer treatment. However, ADMET profiling highlighted that certain properties of strictosidine require further optimization to enhance its drug-like characteristics.

Unlike traditional therapies that target a single protein or chemotherapy that can harm normal cells and cause significant side effects, natural metabolites can affect multiple pathways. This broader approach may improve treatment effectiveness while minimizing unwanted effects. By inhibiting ER α signaling, these metabolites could also influence other important pathways that regulate cell behavior, suggesting they may affect a range of cancer-related processes. Like other plant-based compounds such as isoflavones and resveratrol, these metabolites may offer a more complete strategy for treating hormone-sensitive breast cancer, potentially leading to better outcomes (Sarkar et al. 2009; Thumapati et al. 2025). Thus, the identification of ER α inhibitors from *P. volubilis* presents an opportunity for developing novel therapeutics. As interest in plant-based therapies grows, these findings support their integration into existing treatment approaches. However, the *in silico* results require validation through *in vitro* and *in vivo* studies to confirm efficacy and safety. Further research is needed to clarify their mechanisms of action and potential interactions with existing cancer therapies.

Recent studies have highlighted the importance of ADMET profiling in evaluating the safety and efficacy of plant-based compounds in drug development (Ojuka et al. 2023). In this study, ADMET-based pharmacokinetic profiling was essential for assessing the suitability of selected metabolites for therapeutic use. Most metabolites showed parameters within the range observed for 95% of known drugs, with only minor deviations, similar to the control synthetic drug. These deviations can be addressed through structural optimization or formulation adjustments.

Nearly all metabolites complied with Lipinski's rule of five, but pumiloside and strictosidine showed slight violations that could affect oral bioavailability. Tamoxifen, a widely used and effective selective estrogen receptor modulator (SERM) for breast cancer treatment, violates one of Lipinski's rules, yet remains a clinically preferred option. In contrast, raloxifene, which adheres to all of Lipinski's rules, is less commonly used than tamoxifen, suggesting that factors beyond Lipinski's criteria, such as receptor binding efficiency, metabolism, and pharmacodynamic stability, also play a critical role in therapeutic efficacy. While Lipinski's rule of five serves as a guide to predict oral bioavailability, minor violations do not preclude therapeutic effectiveness, particularly when alternative delivery mechanisms or specific transport pathways are utilized. Rigid adherence to these rules may exclude potentially effective compounds, highlighting the risk of overlooking viable drug candidates that do not fully conform to these criteria (Caminero et al. 2023; Egbert et al. 2019).

ADMET profiling is key to identifying characteristics that may limit a compound's potential as a drug. For example, strictosidine and pumiloside, both with potential for cancer

treatment, showed limitations in oral bioavailability. This insight allows for exploring alternative administration routes or modifications to improve absorption. Overall, ADMET profiling plays a crucial role in optimizing the therapeutic potential of natural compounds.

Molecular dynamics (MD) simulations confirmed that strictosidine and pumiloside maintained stability throughout the simulations, with their complexes with the estrogen receptor remaining consistently stable under physiological conditions. RMSD analysis indicated that these two ligands remained more closely associated with the receptor than other metabolites and the control. RMSF findings suggested that most metabolites did not induce significant conformational changes, further reinforcing the stability of the protein–ligand complexes. The type and number of interactions observed between strictosidine and the receptor, compared to the control and other metabolites, highlighted its stronger binding affinity. This was evident in the 2D interaction visualizations, highlighting its potential as a promising drug candidate. MM–GBSA analysis further supported this, showing that both pumiloside and strictosidine exhibited stronger binding affinities with the lowest binding free energies, surpassing the control drug, tamoxifen. This reflected the enhanced binding strength and stability of their receptor–ligand complexes. While strictosidine outperformed other metabolites, including pumiloside, in terms of structural stability, higher glide energy, and stronger interactions, However, many of the analyzed compounds demonstrated significant potential as drug candidates, each offering advantages in different aspects of molecular interaction and stability. Overall, pumiloside excels in lipophilic and van der Waals interactions, whereas strictosidine demonstrates superior electrostatic and hydrogen bond interactions. Both compounds significantly outperform tamoxifen, suggesting that pumiloside and strictosidine are promising SERMs for breast cancer treatment by inhibiting ER α . Their distinct interaction profiles highlight the potential for further optimization and development as targeted therapeutics (Bake et al. 2025).

Previous studies have confirmed the cytotoxic effects of *P. volubilis* seed extracts against breast cancer cells, with one study showing that treatment with these extracts induced chromatin condensation in MCF-7 breast cancer cell lines, further demonstrating the plant's anticancer activity [52]. In this study, *in silico* analyses of its metabolites highlighted their potential as therapeutic candidates for breast cancer treatment. Future research should focus on optimizing extraction and isolation methods to increase the yield of key metabolites from *P. volubilis*.

Despite plant metabolites therapeutic promise, challenges such as low bioavailability, complex chemical structures, and limited clinical validation remain barriers to their clinical application. To fully harness their potential, future research

should focus on improving bioavailability, refining targeted delivery approaches, and minimizing side effects. Rigorous in vivo studies and clinical trials will be critical to establishing their safety and efficacy. Collaborative efforts across disciplines will play a key role in translating these promising natural compounds into effective clinical therapies, further supporting the integration of plant-derived metabolites into cancer treatment strategies. Enhancing the production of these metabolites through biotechnological methods, such as plant cell cultures or microbial fermentation, can support their therapeutic applications by providing a consistent and sustainable supply. Tissue culture techniques, such as callus and suspension cultures, offer controlled environments for high-quality metabolite production, reducing the need for wild harvesting and promoting sustainability. Additionally, since these metabolites are found in multiple plant species, exploring alternative sources can help prevent overharvesting and ensure long-term availability. Metabolic engineering to optimize biosynthetic pathways could further increase yields and reduce production costs, supporting the commercialization of these compounds for therapeutic use.

Conclusion

The study investigated *Pyrenacantha volubilis* as a potential source of SERM for breast cancer treatment through in silico studies. Key parameters, including docking scores, glide energy, ADMET profiling, MD simulation (RMSD and RMSF), and MM-GBSA analysis, were conducted to evaluate the ability of metabolites to bind with the Estrogen Receptor Alpha (ER α) and form stable complexes. The results showed that all metabolites, except protonated 10-Hydroxycamptothecin, displayed strong potential as SERMs for ER-positive breast cancer. Among them, pumiloside and strictosidine stood out with the best docking scores. Additionally, ten metabolites conformers had docking scores higher than tamoxifen, the standard SERM, while strictosidine also surpassed tamoxifen in glide energy, indicating a stronger and more stable interaction. These findings provide strong evidence that metabolites from *P. volubilis* may effectively inhibit ER α , regulate estrogen signaling, and potentially prevent metastasis. Furthermore, the presence of bioactive compounds with promising SERM activity underscores the therapeutic potential of *P. volubilis*, warranting further in-depth laboratory investigations to validate its efficacy as a candidate for breast cancer treatment.

Enhancing the production of these metabolites through biotechnological methods or tissue culture could further support their therapeutic applications. Since these metabolites are also found in other plant species, exploring alternative sources could help prevent overharvesting and promote sustainability. This study lays a strong foundation for

developing plant-derived breast cancer therapeutics, emphasizing the need for further in vitro and in vivo validation to advance these compounds toward clinical application.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s13205-025-04315-4>.

Acknowledgements The authors would like to express their sincere thanks to the Centre for Computational Informatics Research and Innovation, MCC-MRF Innovation Park and DNA Barcoding Lab, Department of Botany, Madras Christian College for providing the opportunity and support to conduct this research. Their valuable resources and assistance have significantly contributed to the success of this study.

Author contributions Dr. Maya Ammathil Manoharan and Dr. Beutline Malgija conceived and designed the research. Dr. Maya Ammathil Manoharan conducted the investigation, developed the methodology, and wrote the original draft. Dr. Beutline Malgija and Dr. Senthilkumar Umapathy provided validation and review. All authors approved the final manuscript.

Funding This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability Supplementary datasets and supporting materials are included with this manuscript. Additional data related to the study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare that they have no known competing financial or non-financial interests or personal relationships that could have influenced the work reported in this paper.

References

- Adelusi TI, Oyedele A-QK, Boyenle ID, Ogunlana AT, Adeyemi RO, Ukachi CD, Idris MO, Olaoba OT, Adedotun IO, Kolawole OE, Xiaoxing Y, Abdul-Hammed M (2022) Molecular modeling in drug discovery. *Informat Med Unlocked* 29:100880. <https://doi.org/10.1016/j.imu.2022.100880>
- Ahmad S, Singh V, Gautam HK, Raza K (2024) Multisampling-based docking reveals imidazolidinyl urea as a multitargeted inhibitor for lung cancer: an optimisation followed multi-simulation and in-vitro study. *J Biomol Struct Dyn* 42(5):2494–2511. <https://doi.org/10.1080/07391102.2024.2345657>
- Ahmed H, Abdelraheem A, Salem M, Sabry M, Fekry N, Mohamed F, Caprioli G (2023) Suppression of breast cancer: modulation of estrogen receptor and downregulation of gene expression using natural products. *Nat Prod Res* 38(11):1997–2006. <https://doi.org/10.1080/14786419.2023.2232926>
- Ali S, Coombes RC (2000) Estrogen receptor alpha in human breast cancer: occurrence and significance. *J Mammary Gland Biol Neoplasia* 5:271–281. <https://doi.org/10.1023/A:1009594727358>
- Alzahrani AA, Saleh RO, Latypova A, Bokov DO, Kareem AH, Talib HA, Hameed NM, Pramanik A, Alawadi A, Alsalamy A (2024) Therapeutic significance of long noncoding RNAs in estrogen receptor-positive breast cancer. *Cell Biochem Funct* 42(3):e3993
- Anbalagan M, Rowan BG (2015) Estrogen receptor alpha phosphorylation and its functional impact in human breast cancer. *Mol Cell Endocrinol* 418:264–272. <https://doi.org/10.1016/j.mce.2015.01.016>

- Ariazi EA, Ariazi JL, Cordera F, Jordan VC (2006) Estrogen receptors as therapeutic targets in breast cancer. *Curr Top Med Chem* 6(3):181–202
- Arnal JF, Lenfant F, Metivier R, Flouriot G, Henrion D, Adlanmerini M, Fontaine C, Gourdy P, Chambon P, Katzenellenbogen B, Katzenellenbogen J (2017) Membrane and nuclear estrogen receptor alpha actions: from tissue specificity to medical implications. *Physiol Rev* 97(3):1045–1087. <https://doi.org/10.1152/physrev.00025.2016>
- Bader G, Hogue C (2002) Analyzing yeast protein–protein interaction data obtained from different sources. *Nat Biotechnol* 20:991–997. <https://doi.org/10.1038/nbt1002-991>
- Bajorath J (2002) Integration of virtual and high-throughput screening. *Nat Rev Drug Discov* 1:882–894. <https://doi.org/10.1038/nrd941>
- Bake MM, Fofack HMT, Abdelli W, Ntie-Kang F, Bebiga G, Ndassa IM, Vercauteren DP, Mohammad-Salim HA, de Julián-Ortiz JV (2025) Unlocking potential natural products from African sources targeting overexpressed estrogen receptor alpha (ER α) in breast cancer: *In silico* studies. *Comput Struct Biotechnol Rep* 2:100033. <https://doi.org/10.1016/j.csbr.2025.100033>
- Barone I, Brusco L, Fuqua SA (2010) Estrogen receptor mutations and changes in downstream gene expression and signaling. *Clin Cancer Res* 16(10):2702–2708. <https://doi.org/10.1158/1078-0432.CCR-09-1753>
- Caminero Gomes Soares A, Marques Sousa GH, Calil RL, Goulart Trossini GH (2023) Absorption matters: a closer look at popular oral bioavailability rules for drug approvals. *Mol Inf* 42(11):e202300115. <https://doi.org/10.1002/minf.202300115>
- Cao X, Jiang J, Zhang S, Zhu L, Zou J, Diao Y, Xiao W, Shan L, Sun H, Zhang W, Huang J (2013) Discovery of natural estrogen receptor modulators with structure-based virtual screening. *Bioorg Med Chem Lett* 23(11):3329–3333. <https://doi.org/10.1016/j.bmcl.2013.03.105>
- Cersosimo MG, Benarroch EE (2015) Estrogen actions in the nervous system: complexity and clinical implications. *Neurology* 85(3):263–273. <https://doi.org/10.1212/WNL.0000000000001776>
- Clusan L, Le Goff P, Flouriot G, Pakdel F (2021) A closer look at estrogen receptor mutations in breast cancer and their implications for estrogen and antiestrogen responses. *Int J Mol Sci* 22(2):756. <https://doi.org/10.3390/ijms22020756>
- Cummings SR, Eckert S, Krueger KA, Grady D, Powles TJ, Cauley JA, Norton L, Nickelsen T, Bjarnason NH, Morrow M, Lippman ME (1999) The effect of raloxifene on risk of breast cancer in postmenopausal women: results from the MORE randomized trial. *JAMA* 281(23):2189–2197. <https://doi.org/10.1001/jama.281.23.2189>
- Curtaz CJ, Kiesel L, Meybohm P, Wöckel A, Burek M (2022) Anti-hormonal therapy in breast cancer and its effect on the blood-brain barrier. *Cancers (Basel)* 14(20):5132. <https://doi.org/10.3390/cancers14205132>
- Daoud NEH, Borah P, Deb PK, Venugopala KN, Hourani W, Alzweiri M, Bardaweel SK, Tiwari V (2021) ADMET profiling in drug discovery and development: perspectives of in silico, in vitro and integrated approaches. *Curr Drug Metab* 22(7):503–522. <https://doi.org/10.2174/1389200222666210705122913>
- Dimauro I, Grazioli E, Antiozzi C, Duranti G, Arminio A, Mancini A, Greco EA, Caporossi D, Parisi A, Di Luigi L (2021) Estrogen-receptor-positive breast cancer in postmenopausal women: the role of body composition and physical exercise. *Int J Environ Res Public Health* 18(18):9834. <https://doi.org/10.3390/ijerph18189834>
- Dykstra KD, Guo L, Birzin ET, Chan W, Yang YT, Hayes EC, DaSilva CA, Pai LY, Mosley RT, Kraker B, Fitzgerald PM (2007) Estrogen receptor ligands. Part 16: 2-Aryl indoles as highly subtype selective ligands for ER α . *Bioorg Med Chem Lett* 17:2322–2328. <https://doi.org/10.1016/j.bmcl.2007.01.054>
- Egbert M, Whitty A, Keseru GM, Vajda S (2019) Why some targets benefit from beyond rule of five drugs. *J Med Chem* 62(22):10005–10025. <https://doi.org/10.1021/acs.jmedchem.8b01732>
- Etti I, Abdullah R, Hashim NM, Kadir A, Abdul AB, Etti C, Malami I, Waziri P, How CW (2016) Artonin E and structural analogs from *Artocarpus* species abrogates estrogen receptor signaling in breast cancer. *Molecules* 21(7):839. <https://doi.org/10.3390/molecules21070839>
- Ettinger B, Black DM, Mitlak BH, Knickerbocker RK, Nickelsen T, Genant HK, Christiansen C, Delmas PD, Zanchetta JR, Stakkestad J, Glüer CC (1999) Reduction of vertebral fracture risk in postmenopausal women with osteoporosis treated with Raloxifene: results from a 3-year randomized clinical trial. *JAMA* 282(7):637–645. <https://doi.org/10.1001/jama.282.7.637>
- Feinberg EN, Joshi E, Pande VS, Cheng AC (2020) Improvement in ADMET prediction with multitask deep featurization. *J Med Chem* 63(16):8835–8848. <https://doi.org/10.1021/acs.jmedchem.9b02187>
- Fisher B, Dignam J, Bryant J, DeCillis A, Wickerham DL, Wolmark N, Costantino J, Redmond C, Fisher ER, Bowman DM, Deschênes L (1996) Five versus more than five years of tamoxifen therapy for breast cancer patients with negative lymph nodes and estrogen receptor-positive tumors. *JNCI: J Natl Cancer Inst* 88(21):1529–1542. <https://doi.org/10.1093/jnci/88.21.1529>
- Fisher B, Costantino JP, Wickerham DL, Redmond CK, Kavanah M, Cronin WM, Vogel V, Robidoux A, Dimitrov N, Atkins J, Daly M (1998) Tamoxifen for prevention of breast cancer: report of the national surgical adjuvant breast and bowel project P-1 study. *JNCI: J Natl Cancer Inst* 90:1371–1388. <https://doi.org/10.1093/jnci/90.18.1371>
- Girgert R, Emons G, Gründker C (2019) Estrogen signaling in ER α -negative breast cancer: ER β and GPER. *Front Endocrinol* 9:781. <https://doi.org/10.3389/fendo.2018.00781>
- Gowtham HG, Revanasiddappa PD, Murali M, Singh SB, Abhilash MR, Pradeep S, Shivamallu C, Achar RR, Silina E, Stupin V, Manturova N (2024) Secondary metabolites of *Trichoderma* spp. as EGFR tyrosine kinase inhibitors: evaluation of anti-cancer efficacy through computational approach. *PLoS ONE* 19(1):e0296010. <https://doi.org/10.1371/journal.pone.0296010>
- Jensen EV, Jacobson HI (1960) Fate of steroid estrogens in target tissues. In G. Pincus & E. P. Vollmer (Eds.), *Biological Activities of Steroids in Relation to Cancer*, 161–178. Academic Press. <https://doi.org/10.1016/B978-1-4832-2866-2.50015-5>
- Jordan VC (2003) Tamoxifen: a most unlikely pioneering medicine. *Nat Rev Drug Discov* 2:205–213. <https://doi.org/10.1038/nrd1031>
- Karunaratna I, Aluthge P, Gunawardana K, Rajapaksha S, De Alvis K, Gunathilake ST, Hapuarachchi U, Gunasena P, Fernando C, Warnakulasooriya A, Perera N, Tenney S, Obboh-Weilke A, Wagner D, Chen MY (2024) Tamoxifen in breast cancer therapy: clinical applications and outcomes. *Survey Ophthalmol* 69(1):42–50. <https://doi.org/10.1016/j.survophthal.2024.01.003>
- Kim C, Kim B (2018) Anti-cancer natural products and their bioactive compounds inducing ER stress-mediated apoptosis: a review. *Nutrients* 10(8):1021. <https://doi.org/10.3390/nu10081021>
- Lee J, Kim J, Jeong C, Baek KH, Ha J (2024) Beyond breast cancer: role of selective estrogen receptor modulators in reducing systemic malignancies: evidence from population-based data. *Curr Med Res Opin* 40(9):1589–1596. <https://doi.org/10.1080/03007995.2024.2390649>
- Li JWH, Vederas JC (2009) Drug discovery and natural products: end of an era or an endless frontier? *Science* 325(5937):161–165. <https://doi.org/10.1126/science.1168243>

- Lipinski CA (2004) Lead-and drug-like compounds: the rule-of-five revolution. *Drug Discov Today: Technol* 1(4):337–341. <https://doi.org/10.1016/j.ddtec.2004.11.007>
- Lumachi F, Brunello A, Maruzzo M, Basso U, Basso SM (2013) Treatment of estrogen receptor-positive breast cancer. *Curr Med Chem* 20(5):596–604. <https://doi.org/10.2174/092986713804999303>
- MacQueen J (1967) Some methods for classification and analysis of multivariate observations. *Proceedings of the fifth berkeley symposium on mathematical statistics and probability, volume 1: statistics*. University of California Press, pp 281–298
- McInerney EM, Tsai MJ, O'Malley BW, Katzenellenbogen BS (1996) Analysis of estrogen receptor transcriptional enhancement by a nuclear hormone receptor coactivator. *Proc Natl Acad Sci* 93(19):10069–10073. <https://doi.org/10.1073/pnas.93.19.10069>
- Mishra D, Mishra A, Nand Rai S, Vamanu E, Singh MP (2023a) Demystifying the role of prognostic biomarkers in breast cancer through integrated transcriptome and pathway enrichment analyses. *Diagnostics (Basel)* 13(6):1142. <https://doi.org/10.3390/diagnosics13061142>
- Mishra D, Mishra A, Rai SN, Singh SK, Vamanu E, Singh MP (2023b) In silico insight to identify potential inhibitors of BUB1B from mushroom bioactive compounds to prevent breast cancer metastasis. *Front Biosci (Landmark Edition)* 28(7):151. <https://doi.org/10.31083/j.fbl2807151>
- Muccee F, Ashraf NM, Razak S, Afsar T, Hussain N, Husain FM, Shafique H (2024) Exploring the association of ESR1 and ESR2 gene SNPs with polycystic ovary syndrome in human females: a comprehensive association study. *J Ovarian Res* 17(1):27. <https://doi.org/10.1186/s13048-023-01335-7>
- Nagandla H, Thomas C (2024) Estrogen signals through ER β in breast cancer; what we have learned since the discovery of the receptor. *Receptors* 3(2):182–200. <https://doi.org/10.3390/receptors3020010>
- Ojuka P, Kimani NM, Apollo S, Nyariki J, Ramos RS, Santos CB (2023) Phytochemistry of plants in the genus *Vepris*: a review and in silico analysis of their ADMET properties. *South Afr J Bot* 157:106–114. <https://doi.org/10.1016/j.sajb.2023.03.057>
- Pantaleao SQ, Fernandes PO, Gonçalves JE, Maltarollo VG, Honorio KM (2022) Recent advances in the prediction of pharmacokinetics properties in drug design studies: a review. *ChemMedChem*. <https://doi.org/10.1002/cmde.202100542>
- Roy R, Moses MA (2012) ADAM12 induces estrogen-independence in breast cancer cells. *Breast Cancer Res Treat* 131:731–741. <https://doi.org/10.1007/s10549-011-1431-4>
- Saczko J, Michel O, Chwiłkowska A, Sawicka E, Mączynska J, Kulbacka J (2017) Estrogen receptors in cell membranes regulation and signaling. In: Kulbacka J, Satkauskas S (eds) *Transport across natural and modified biological membranes and its implications in physiology and therapy*. Springer, Cham, pp 93–120
- Sarkar FH, Li Y, Wang Z, Kong D (2009) Cellular signaling perturbation by natural products. *Cellular Signal* 21(11):1541–1547. <https://doi.org/10.1016/j.cellsig.2009.03.009>
- Schrodinger LLC (2024) Docking and scoring: The role of molecular docking in structure-based drug design. Retrieved October 27, 2024, from <https://www.schrodinger.com/life-science/learn/white-papers/docking-and-scoring>.
- Sharma D, Kumar S, Narasimhan B (2018) Estrogen alpha receptor antagonists for the treatment of breast cancer: a review. *Chem Cent J* 12:107. <https://doi.org/10.1186/s13065-018-0472-8>
- Suma HK, Kumar V, Senthilkumar U, Kumara PM, Ravikanth G, Santhoshkumar TR, Shaanker RU (2014) *Pyrenacantha volubilis* Wight (Icacinaeae) a rich source of camptothecine and its derivatives, from the coromandel coast forests of India. *Fitoterapia* 97:105–110. <https://doi.org/10.1016/j.fitote.2014.05.017>
- Szklarczyk D, Gable AL, Lyon D, Junge A, Wyder S, Huerta-Cepas J, Simonovic M, Doncheva NT, Morris JH, Bork P, Jensen LJ (2019) STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Res*. <https://doi.org/10.1093/nar/gky1131>
- Tecalco-Cruz AC, Ramírez-Jarquín JO, Cruz-Ramos E (2019) Estrogen receptor alpha and its ubiquitination in breast cancer cells. *Curr Drug Targets* 20(6):690–704. <https://doi.org/10.2174/1389450119666181015114041>
- Thumpati P, Rai SN, Prajapati C, Ramakrishna K, Singh SK (2025) Targeting c-MYC G-quadruplexes for cancer treatment with small molecules. *Sci Pharm* 93(1):6. <https://doi.org/10.3390/scipharm93010006>
- Tie H, Walker BD, Singleton CB, Valenzuela SM, Bursill JA, Wyse KR, Breit SN, Campbell TJ (2009) Inhibition of HERG potassium channels by the antimalarial agent halofantrine. *Br J Pharmacol*. <https://doi.org/10.1038/sj.bjp.0703470>
- Valera MC, Fontaine C, Dupuis M, Noirrit-Esclassan E, Vinel A, Guillaume M, Gourdy P, Lenfant F, Arnal JF (2018) Towards optimization of estrogen receptor modulation in medicine. *Pharmacol Ther* 189:123–129. <https://doi.org/10.1016/j.pharmthera.2018.05.002>
- Weatherman RV, Fletterick RJ, Scanlan TS (1999) Nuclear-receptor ligands and ligand-binding domains. *Annu Rev Biochem* 68(1):559–581. <https://doi.org/10.1146/annurev.biochem.68.1.559>
- Xu R, Yang X, Tang B, Mao Y, Jiang F (2024) Combined treatment of all-trans retinoic acid with Tamoxifen suppresses ovarian cancer. *Cancer Chemother Pharmacol* 94:259–270. <https://doi.org/10.1007/s00280-024-04671-7>
- Yao S, Zhu Y, Chen L (2013) Advances in targeting cell surface signalling molecules for immune modulation. *Nat Rev Drug Discov* 12:130–146. <https://doi.org/10.1038/nrd3877>

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.